



UNIVERSITY OF
ALBERTA

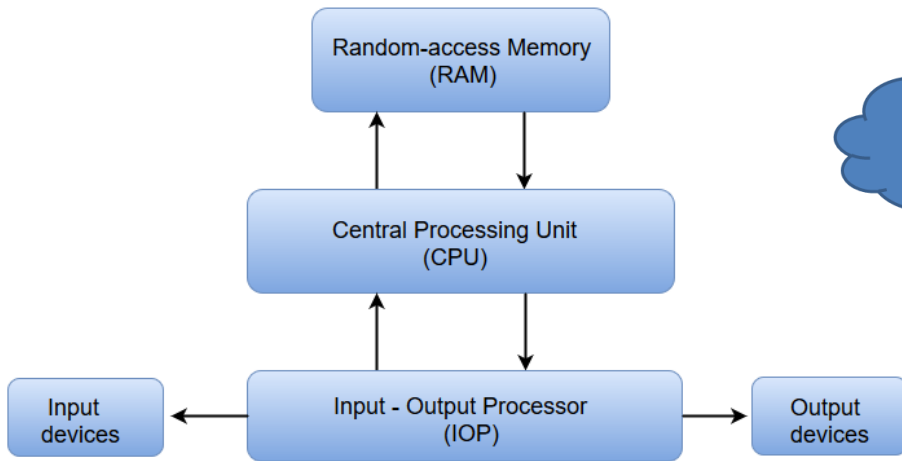
Course Overview

CMPUT 382

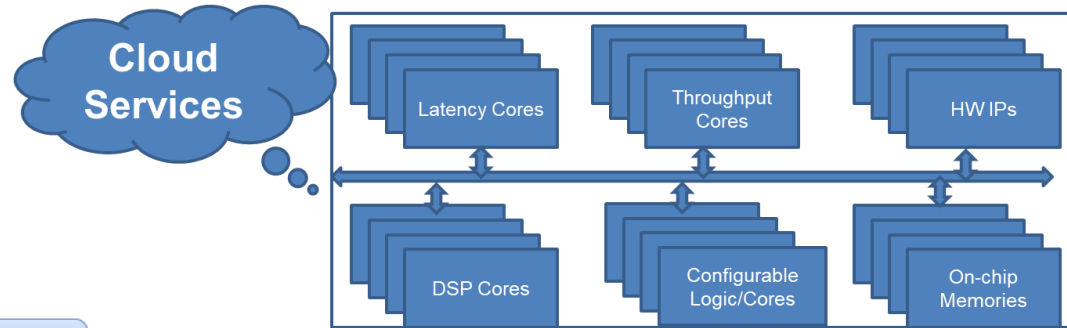
Introduction to GPU Programming

Pierre Boulanger

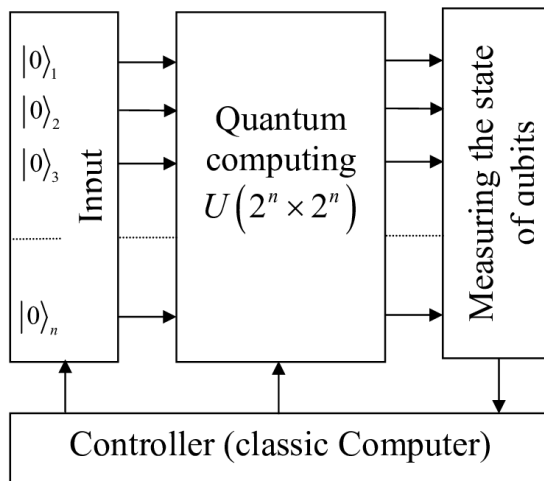
Various Computing Paradigms



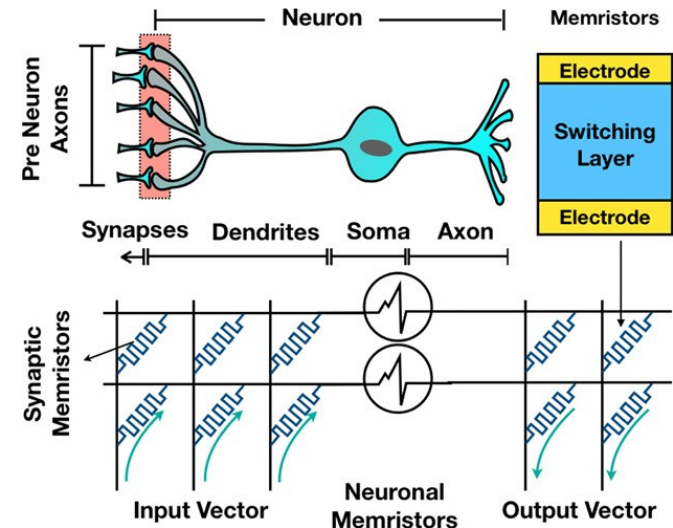
Digital Computer



Heterogenous Digital Computer



Quantum Computer



Nuromorphic Computer

What is a Heterogeneous System



Heterogeneous, parallel computing
With SYCL

**Why do we need
Heterogeneous Systems?**



<https://www.youtube.com/watch?app=desktop&v=kqj4VlwXk9c>

What this Course is About?

- Learn how to program heterogeneous parallel computing systems and achieve
 - High performance and energy-efficiency
 - Functionality and maintainability
 - Scalability across future generations
 - Portability across vendor devices
- Technical subjects
 - Parallel programming API, tools and techniques
 - CUDA Language
 - Principles and patterns of parallel algorithms
 - Processor architecture features and constraints

General Information

- Website
 - <https://webdocs.cs.ualberta.ca/~pierreb/CMPUT382>
 - Username: cmput382
 - Password: vizroom
- Instructor
 - Pierre Boulanger
 - Office: 411 Athabasca Hall (by appointment)
 - Email: pierreb@ualberta.ca
- TAs
 - Vishwajeet Ohal ohal@ualberta.ca
 - Ali Parsaee parsaee@ualberta.ca

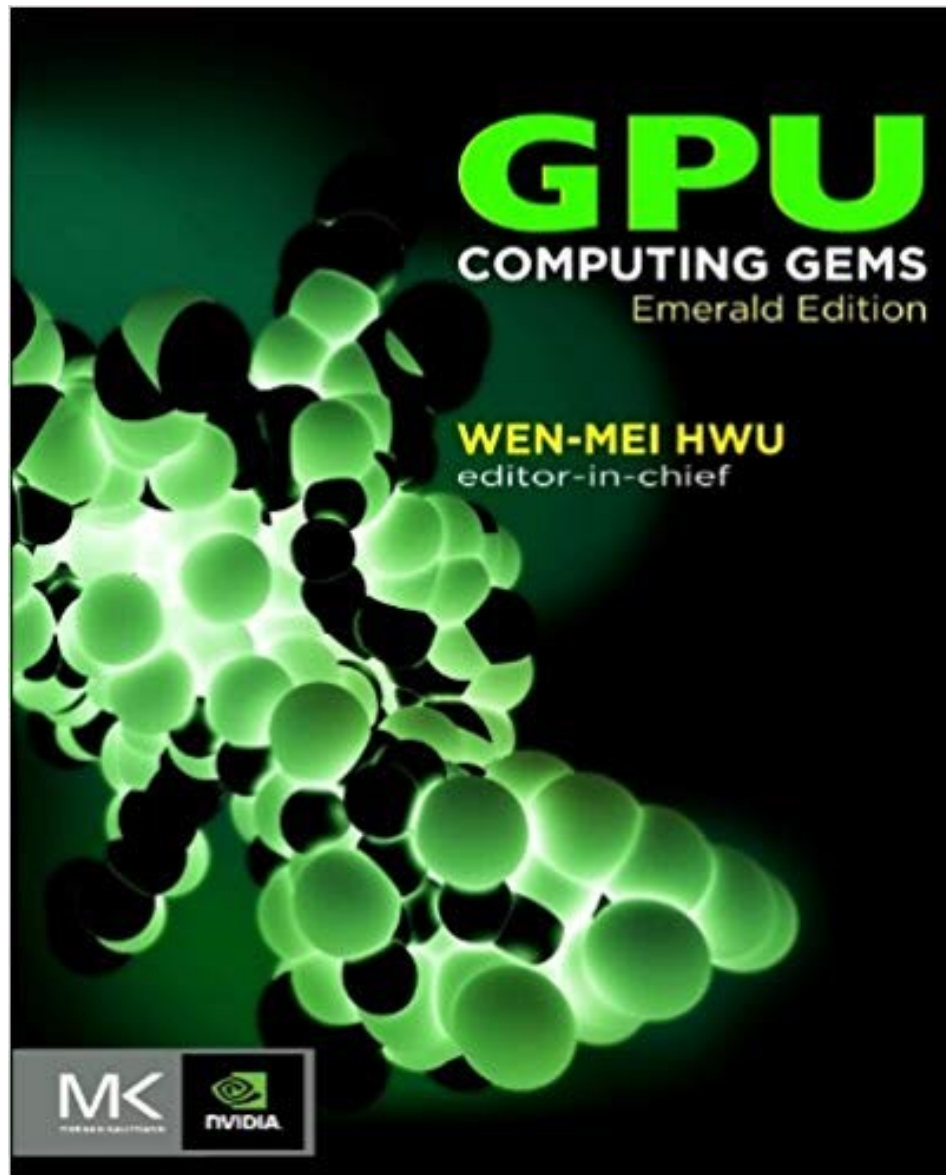
Class Delivery

- Virtual Class
 - Lectures will be done online using Zoom
 - Register in advance for this meeting:
- All lectures will be recorded and made available in e-class that day

Course Materials

- Lecture Notes
 - The lecture notes will be **published on the eClass website**. They are copyright protected, and you are not permitted to pass them on to others
 - This course refers to **GPU Computing Gem Book**, but an electronic version of the chapters is provided with the course
 - Pointers to resources will be made available on the course website or e-Class
 - **All lectures will be delivered using Zoom**
 - **All lectures will be recorded and made available on e-class**

Recommended Reference Book



Course Components

- Laboratories
 - 50%
- Final Exam
 - 30%
- Quizzes
 - 20%

Laboratories

- There will be **10 labs** of various complexity distributed during the term
- The labs will count for 50% of the final score
- The lab instructions will be distributed on the website at the end of each module
- The students are not expected to do the entire lab in the allocated lab period and that they can work from home or when ever the lab is free
- The marks will be based on functionality of the program and speed-up (10% extra)

Google Colab

- Google Colab is a free cloud service
- Colab offers GPU and is completely free
- With Colab you can work on the GPU with CUDA C/C++ for free
- This year, all labs will be using Google Colab using Jupyter Notebook
- <https://colab.research.google.com/>

Google Colab

- Run CUDA C/C++ on Jupyter notebook in Google Colaboratory
- <https://www.geeksforgeeks.org/how-to-run-cuda-c-c-on-jupyter-notebook-in-google-colaboratory/>

Lab Sections With TAs

- 66 students are registered in the class
- For the labs, you have two options
 - Go to your lab section, and you will have access to a computer, Colab, and your TA
 - Simply used Colab at home to compile and execute your CUDA code
- Find out which lab you are registered in:
 - LAB D01 T 14:00 - 16:50 (CSC 1-67)
 - LAB D02 T 17:00 - 19:50 (CSC 1-67)
 - LAB D03 H 14:00 - 16:50 (CSC 1-67)

Quizzes

- There will be 12 quizzes distributed after each module
- You are required to do only 10 of them and it will count for 20% of the final mark
- The quizzes will generally be handed out electronically after a module is terminated and are due in class at the beginning of the next module

Deadlines

- All course components must be **submitted through e-class**
- All course components are due on the deadline date at 12 am except for quizzes which are due on every Monday at 23h55
- **There will be no extensions to deadlines**

Final Exam

- At the end of the term there will be a 2 hours final exam in the form of a quiz
- The Quiz will be administered Using ExamLock
- Exam date: TBA
- The exam will cover the material discussed in class
- The exam will be very close to the quizzes
- The exam will be open book
- The exam will count for 30% of the final mark

Academic Integrity

- The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect
- Students are particularly urged to familiarize themselves with the provisions of the Code of Student Behavior (www.uofaweb.ualberta.ca/gfcpolicymanual/) and avoid any behavior which could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts and/or participation in an offence
- **Academic dishonesty is a serious offence and can result in suspension or expulsion from the University** (GFC 29 SEP 2003)

Academic Integrity

- Detection of plagiarism is easy and fast using tools like Moss
- Everybody's assignment is compared to everybody else's, even over different terms, if necessary
- **All cases of plagiarism will be reported to the Dean's office**
- You are equally guilty whether you took someone else's work, or you gave your work to someone else

GPU Computing in Many Disciplines

**Financial
Analysis**

**Scientific
Simulation**

**Engineering
Simulation**

**Data
Intensive
Analytics**

**Medical
Imaging**

**Digital Audio
Processing**

**Digital Video
Processing**

**Computer
Vision**

**Biomedical
Informatics**

**Electronic
Design
Automation**

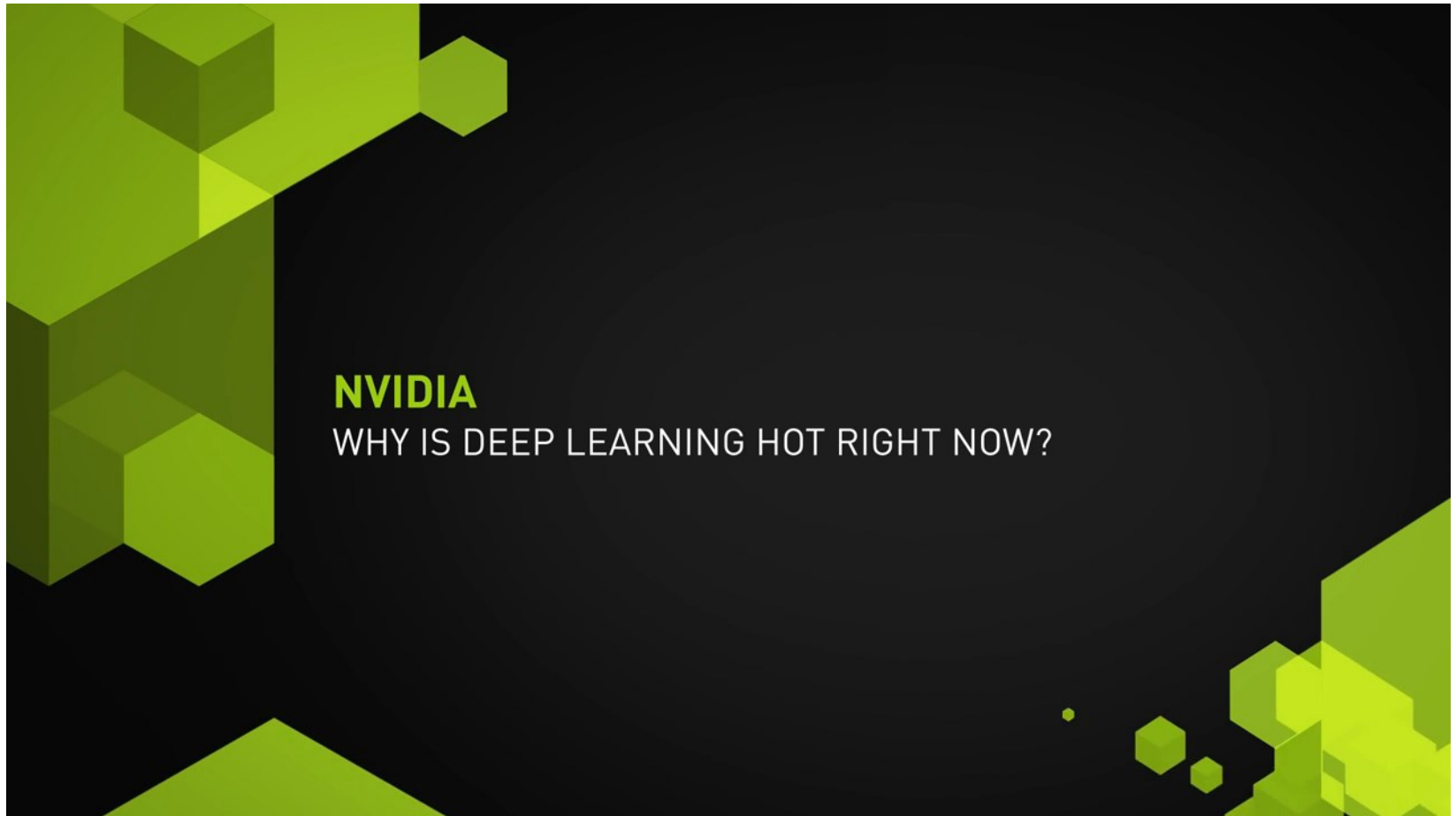
**Statistical
Modeling**

**Numerical
Methods**

**Ray Tracing
Rendering**

**Interactive
Physics**

GPU for Deep Learning



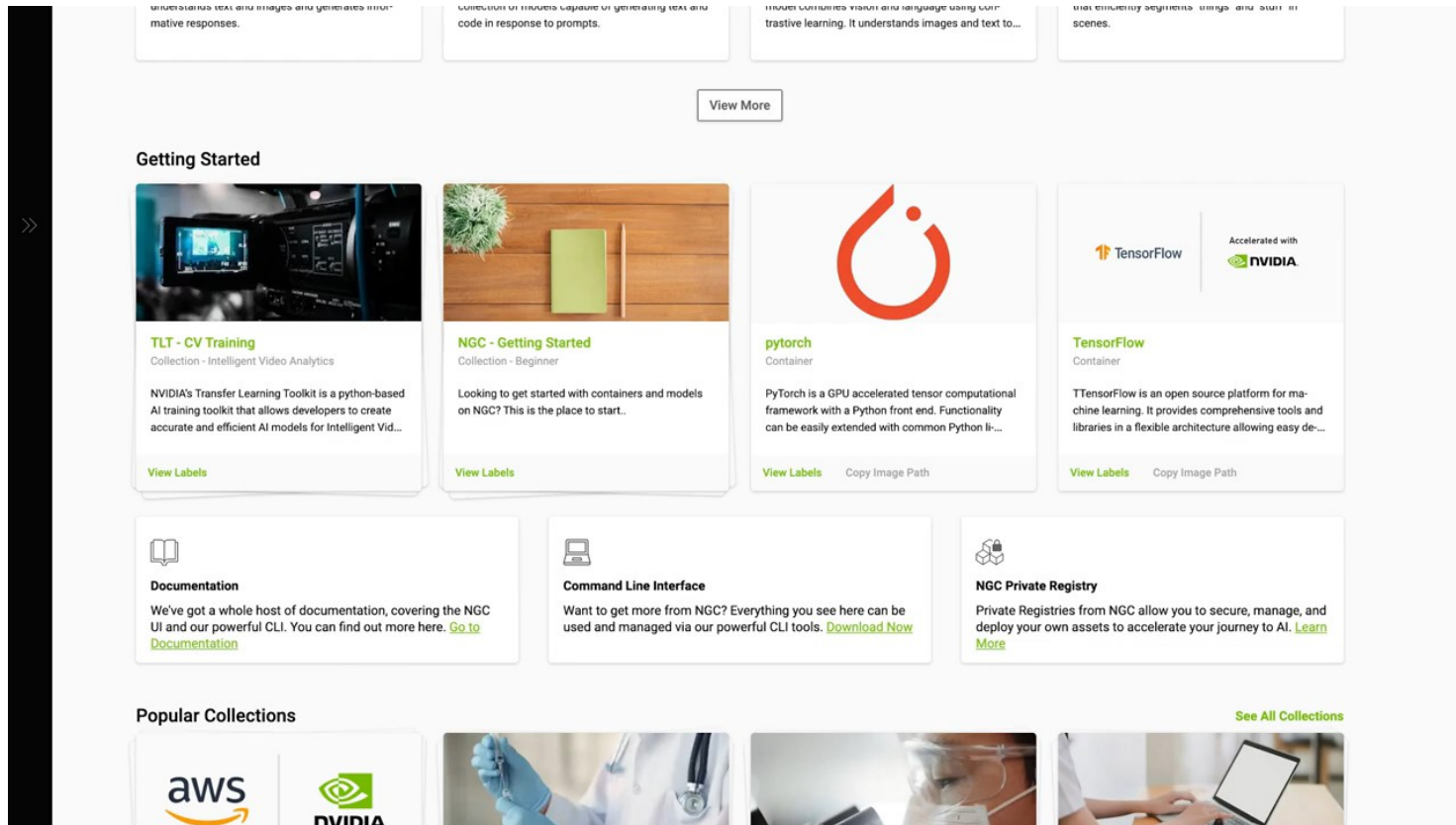
<https://youtu.be/1G0e-mR9a4k>

Omniverse BMW Virtual Factory



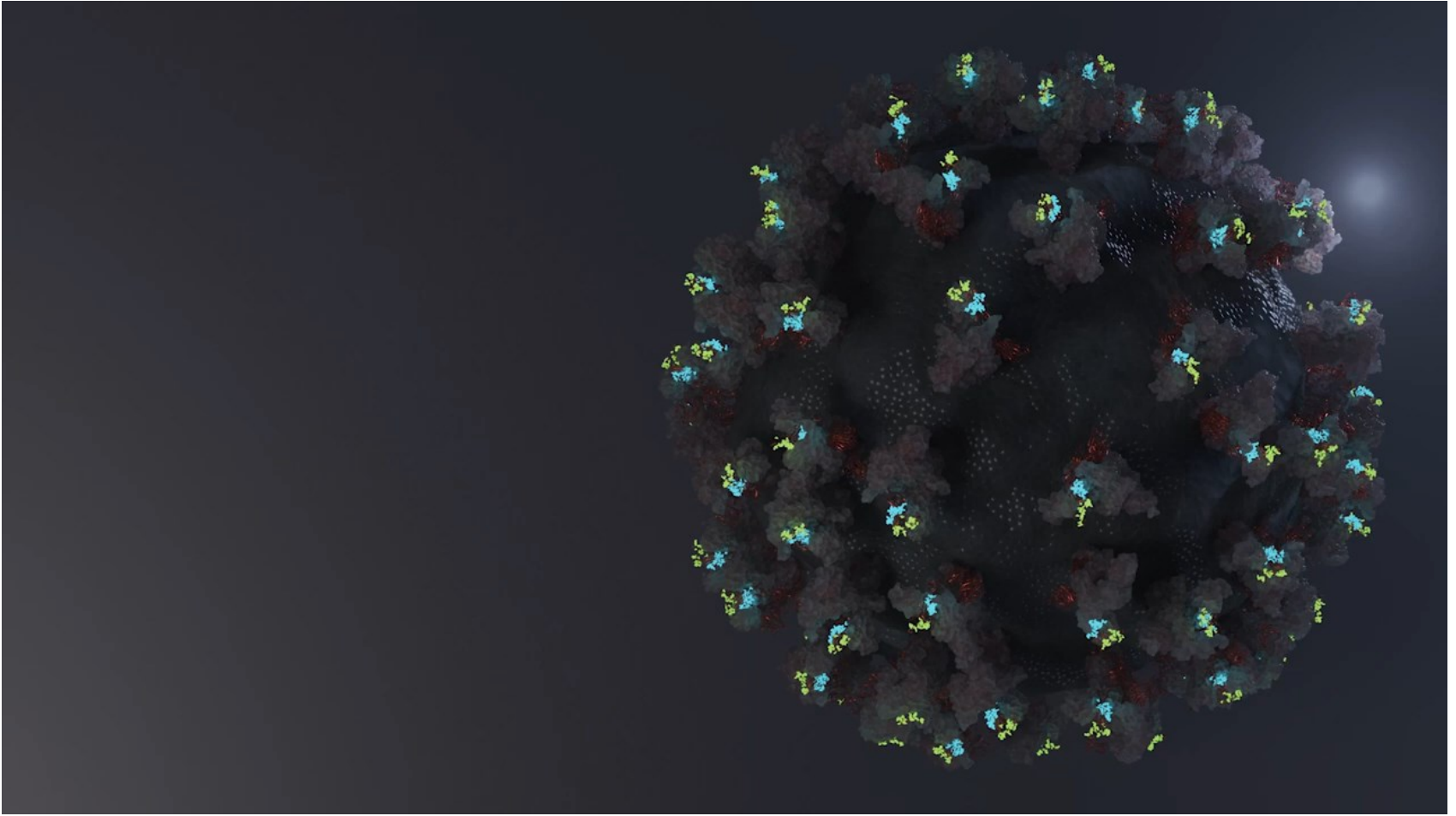
<https://www.youtube.com/watch?v=g78YHYXXils&t=5s>

Nvidia Generative AI on Desktop



<https://www.youtube.com/watch?v=gjoWSRoFEhQ>

Visualizing COVID 19 Virus



<https://www.youtube.com/watch?v=jiMZYJ--cT8&t=7s>

GPU for Molecular Biology



<https://youtu.be/D3fOXt4MrOM>